LAB 9 - Write the RC4 logic in Java Using Java Cryptography, encrypt text "Hello world" using Blowfish Create your own key using Java key tool.

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;
import javax.xml.bind.DatatypeConverter;

public class lab9rc4 {
    public static void main(String[] args) throws Exception {

        String message = "Hello world";

        // Generate a 128-bit RC4 key
        KeyGenerator keyGenerator = KeyGenerator.getInstance("RC4");
        keyGenerator.init(128);

        SecretKey key = keyGenerator.generateKey();

        // Create RC4 cipher
        Cipher cipher = Cipher.getInstance("RC4");

        // Encrypt the message
        cipher.init(Cipher.ENCRYPT_MODE, key);
```

```
        byte[] encrypted =
                cipher.doFinal(message.getBytes("UTF-8"));

        // Decrypt the message
        cipher.init(Cipher.DECRYPT_MODE, key);

        byte[] decrypted =
                cipher.doFinal(encrypted);

        System.out.println("Original Text  : " + message);

        System.out.println("Secret Key     : " +
                DatatypeConverter.printBase64Binary(
                        key.getEncoded()));

        System.out.println("Encrypted Text : " +
                DatatypeConverter.printBase64Binary(
                        encrypted));

        System.out.println("Decrypted Text : " +
                new String(decrypted, "UTF-8"));
    }
}
```